

NOVA RETAIL

# Profitability Optimization Report

*How can the company grow profit without relying only on higher sales?*

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July 2026

# Executive Summary

**Nova Retail is profitable, but the profit is not evenly distributed.** Some products, markets and discounts create strong profit, while others reduce the company result. The main business opportunity is to protect margin, not simply sell more.

|                 |                |               |                   |
|-----------------|----------------|---------------|-------------------|
| <b>\$12.64M</b> | <b>\$1.47M</b> | <b>11.61%</b> | <b>24.46%</b>     |
| Total Sales     | Total Profit   | Profit Margin | Loss-Making Lines |

**Main conclusion:** sales increased strongly from 2011 to 2014, but almost one quarter of order lines still lose money. The largest risks are extreme discounts, the Tables sub-category, weak-margin markets and high shipping costs on faster delivery modes.

## In simple words

The company is making money overall, but many individual sales are not healthy. Better pricing and product decisions can improve profit even if sales stay at the same level.

## Simple definitions

**Profit:** the money left after the cost of the sale.

**Profit margin:** profit as a percentage of sales. A higher margin means the company keeps more money from each sale.

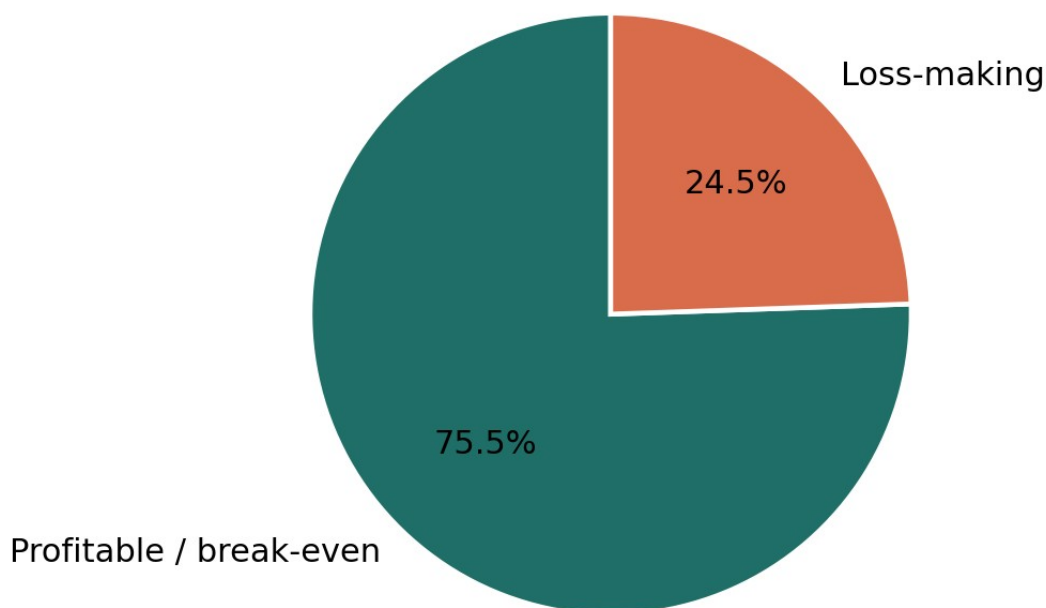
**Loss-making order line:** a product sale where profit is below zero.

**Discount band:** a group of sales with a similar discount level.

# 1. Overall Business Performance

The analysis contains 51,290 order lines, 25,035 orders and 4,873 customers. Total sales are \$12.64 million and total profit is \$1.47 million.

Almost one in four order lines loses money



*Figure 1. Share of profitable and loss-making order lines.*

**Key risk:** 12,544 order lines lose money. This is 24.46% of all lines, so the total profit hides a large number of weak transactions.

## What this means

Managers should not look only at total sales and total profit. They should also track how many individual sales lose money and why.

## 2. Sales and Profit Growth

Sales and profit increased every year. This is positive, but the profit margin stayed close to 11-12%, so the company did not become much more efficient.

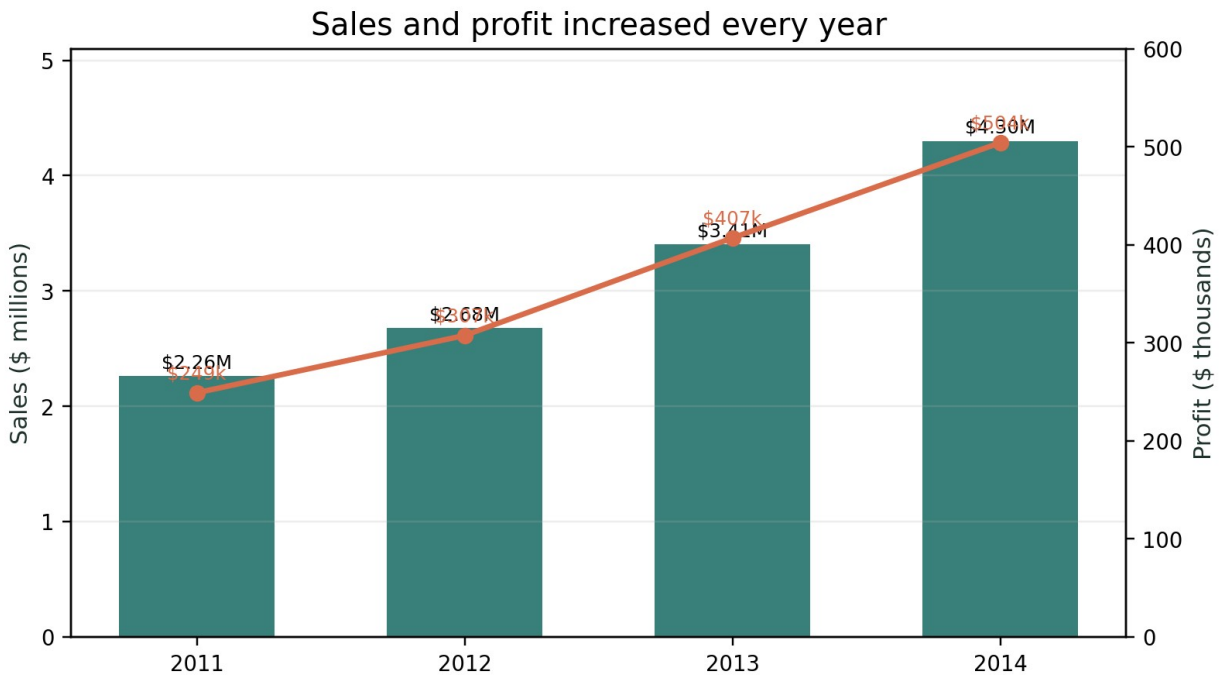


Figure 2. Yearly sales and profit from 2011 to 2014.

**Growth result:** sales increased from \$2.26M in 2011 to \$4.30M in 2014. Profit increased from \$249k to \$504k. However, the margin moved only from about 11.0% to 11.7%.

### What this means

The company is growing mainly because it sells more. The next step should be to earn more profit from each dollar of sales.

### 3. Product and Category Performance

Technology is the strongest category. Furniture generates high sales, but it keeps much less profit from those sales.

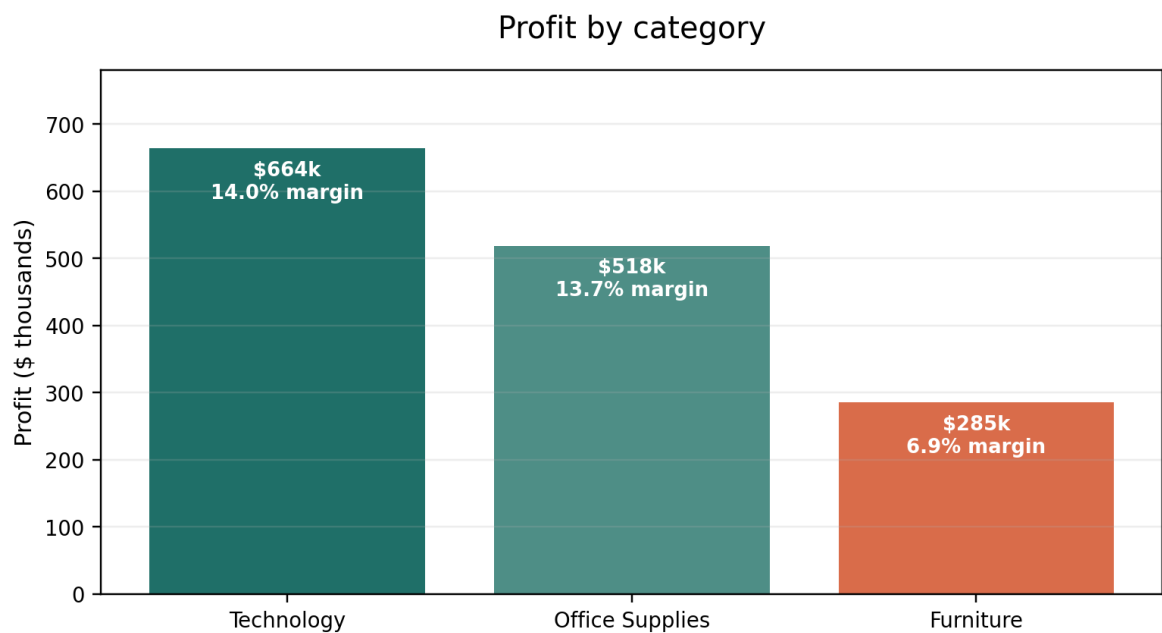


Figure 3. Total profit and margin by category.

**Category message**

Technology and Office Supplies both keep around 14% of sales as profit. Furniture keeps only about 7%, so it needs tighter pricing and cost control.

### 3.1 Sub-category focus

The sub-category view shows which product groups support profit and which ones reduce it.

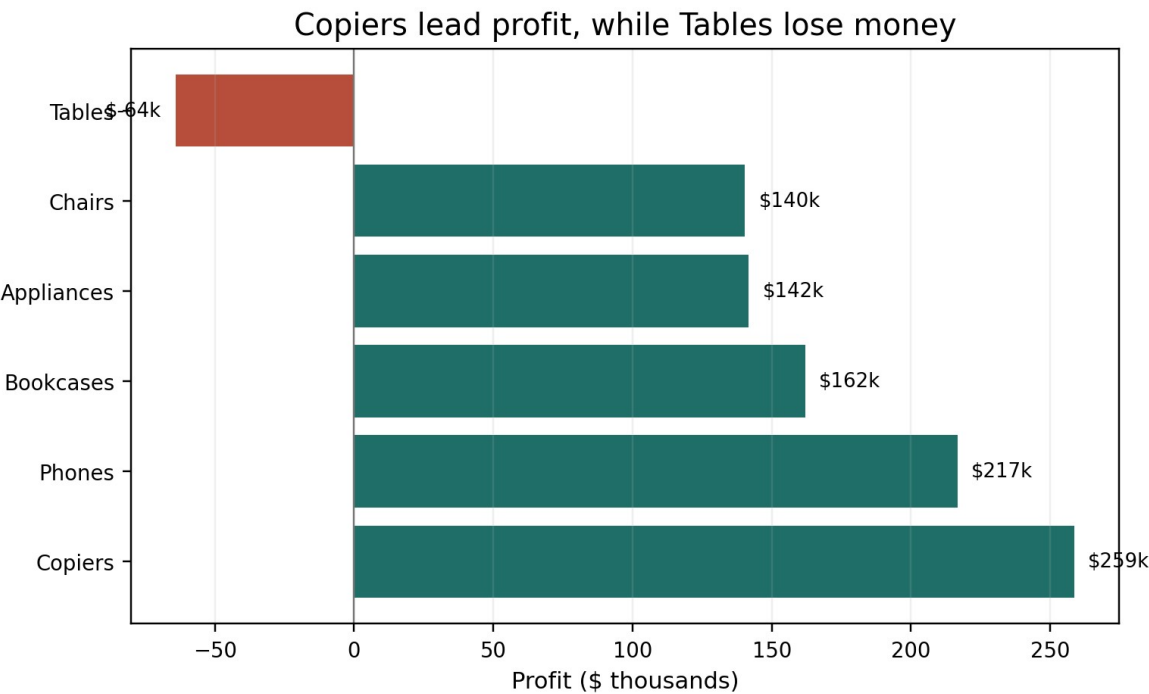


Figure 4. Selected sub-categories by total profit.

**Main product risk:** Tables generated about \$757k in sales but lost about \$64k. This shows that strong sales do not always mean a healthy product line.

**What this means**

Technology should continue to receive support. Tables need a detailed review of prices, discounts, suppliers and shipping costs. Products should be judged by profit and margin, not sales alone.

## 4. Market Performance

APAC, EU and US generate most of the company profit. EMEA is the clearest margin concern.

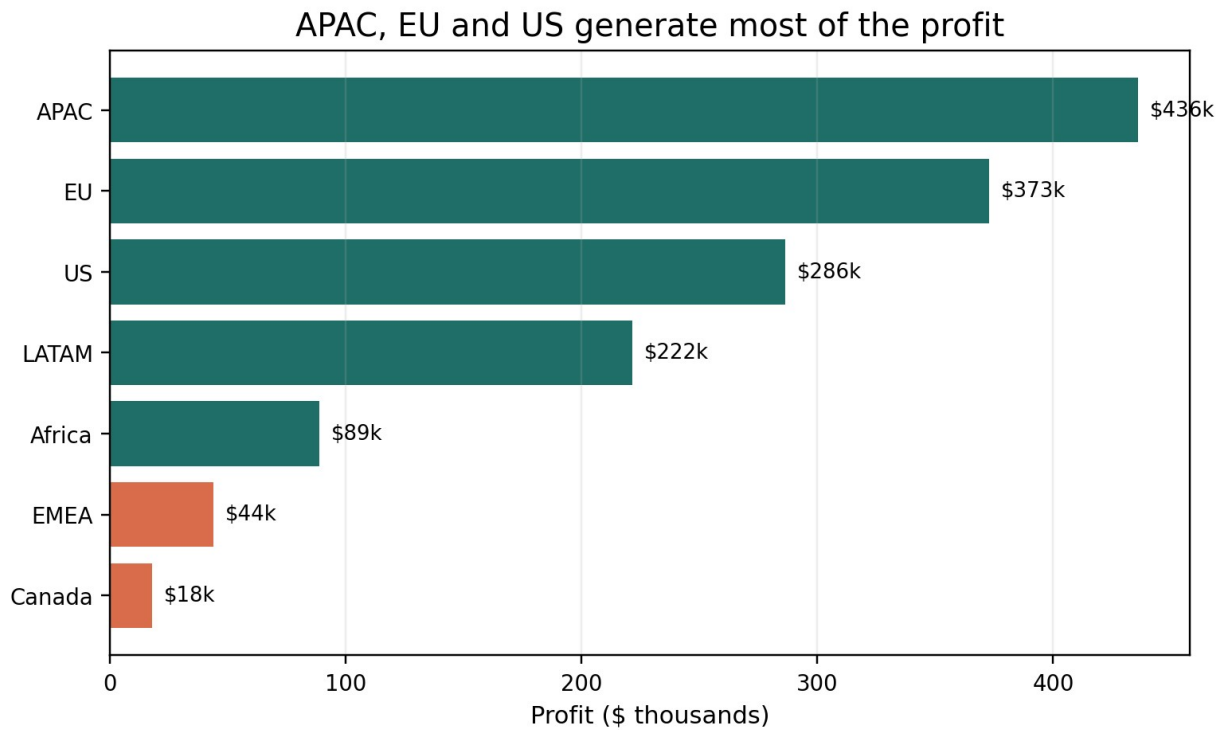


Figure 5. Total profit by market.

**Important detail:** Canada has the lowest total profit because it is a very small market, but its margin is high. EMEA is more worrying because it has larger sales but only about 5% margin.

### What this means

Continue investing in APAC, EU and US, but monitor margin. Review the product mix, prices and discounts in EMEA before trying to increase its sales volume.

# 5. Discount Impact

Discounts are the clearest driver of poor profitability in this project. The difference between no discount and extreme discount is very large.



Figure 6. Profit, margin and loss rate at the two ends of the discount scale.

**No discount:** about \$1.77M profit, a 25% margin and no loss-making rows in this band.

**Extreme discount:** about \$413k loss, a -111% margin and a 100% loss rate.

**What this means**  
Extreme discounts are not a growth tool in this dataset; they destroy profit. Nova Retail should set maximum discount limits by category, product and market.

## 6. Customer Segments and Shipping

The Consumer segment creates the most total profit because it is the largest segment. Home Office has the best margin, but on a smaller sales base.

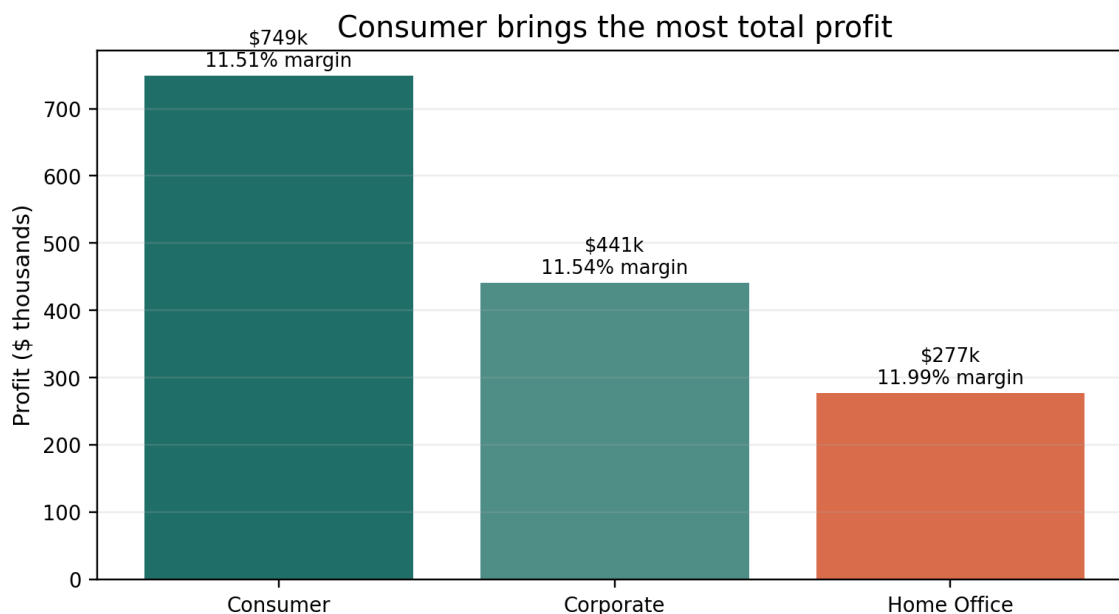


Figure 7. Total profit and margin by customer segment.

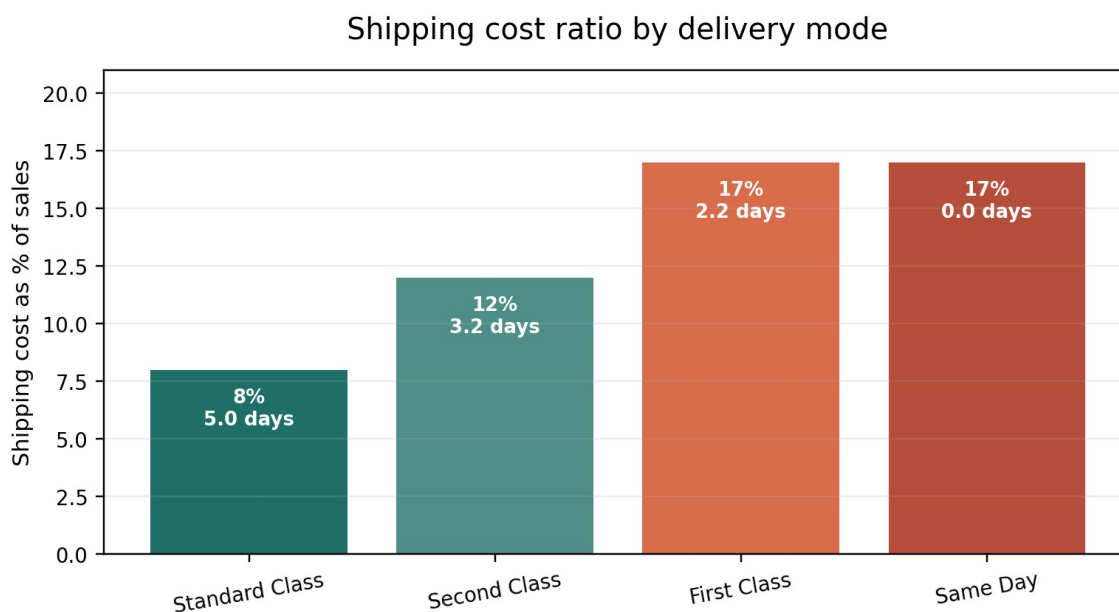


Figure 8. Shipping cost ratio and average delivery time.

**Shipping insight:** Standard Class is slower, but shipping cost is only about 8% of sales. First Class and Same Day use about 17% of sales, while profit margins remain similar.

### What this means

Use faster delivery when the customer pays for it or when the order margin can support it. Do not give expensive shipping options automatically on low-margin orders.

## 7. Business Recommendations

### 1. Control discount levels

Create discount limits by category and market. Extreme discounts should require approval.

### 2. Fix the Tables sub-category

Review prices, supplier cost, shipping cost and the products that create the largest losses.

### 3. Manage markets by both profit and margin

Protect APAC, EU and US. Review EMEA because its margin is much weaker.

### 4. Protect shipping margin

Use Standard Class as the default when speed is not essential. Charge for faster service when needed.

### 5. Track loss-making order lines

Add a regular KPI for loss rate, not only total sales and total profit.

### 6. Focus on profitable growth

Support Technology and strong products, but avoid increasing volume in weak-margin areas.

### Recommended priority order

- First: stop or approve extreme discounts.
- Second: investigate Tables and the biggest loss-making products.
- Third: review EMEA and expensive shipping choices.
- Fourth: build regular monitoring for loss rate and margin.

## 8. Analysis Limitations

### Important context

- The data covers 2011 to 2014, so it is useful for analysis practice but not for current retail decisions.
- The dataset shows what happened, but it does not explain every business cause.
- A negative relationship between discount and profit does not prove that every discount is bad. The right level depends on product cost, customer behaviour and strategy.
- Market totals can be affected by market size. Total profit and margin should always be read together.

### How to use these findings

- Use the results to guide pricing, discount and product reviews.
- Test each change and monitor profit, margin and loss rate over time.
- Use current cost and customer data before applying the same rules to future decisions.

## 9. Final Conclusion

**Nova Retail does not mainly have a sales problem. It has a profit quality problem.** The business is growing and profitable, but a large share of transactions lose money. The strongest improvement opportunities are better discount control, product-level action on Tables, market margin reviews and more careful shipping choices.

### Final message

Nova Retail can improve profit without depending only on more sales. The company should keep more profit from the sales it already makes.

### Project source

[GitHub - Profitability Optimization](#)

**Report basis:** the project README, Python notebooks, SQL business analysis and exported performance tables in the repository. The charts in this report were rebuilt from the project results for a clear executive presentation.